

DATA 485 | Capstone Project Report

Predicting Game Revenue on Steam: What Drives Financial Success Among Top-Selling Titles?

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Spring 2026

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1. Introduction

1.1 The Price of Play: Analyzing Revenue Performance in the Steam Marketplace

The video game industry has become one of the most important and profitable sectors of digital entertainment. Games are now a global business shaped by software development, consumer behavior, digital storefronts, pricing strategies, reviews, online communities, and long-term player engagement. As the market grows more competitive, understanding what is associated with game revenue has become valuable for developers, publishers, analysts, investors, and platform stakeholders.

Steam is especially important because it is one of the largest and most influential storefronts for PC gaming. It serves not only as a sales platform, but also as an ecosystem for discovery, recommendations, visibility, and community feedback. For many PC games, Steam performance can strongly influence audience reach and commercial success.

This capstone project uses Gamalytic Steam analytics data to examine revenue performance patterns across Steam games. The dataset includes estimated revenue and sales, pricing, reviews, wishlist activity, average playtime, publisher class, price tier, and review tier. The project uses data cleaning, transformation, descriptive statistics, statistical testing, regression, and Random Forest modeling to identify patterns associated with revenue performance.

Because game revenue is rarely explained by one factor alone, the analysis compares revenue with multiple market and performance signals. These include price, review score, review count, wishlist activity, average playtime, publisher class, price tier, and review tier. Direct revenue-linked fields such as copies sold are used for descriptive context but excluded from predictive modeling to avoid data leakage.

Overall, this project uses Steam performance data to better understand the business side of gaming. The final analysis and dashboard are designed to make revenue patterns easier to see and explain which factors are most strongly associated with commercial performance in the Steam marketplace.

2. Data Description

2.1 Data Source

The dataset was sourced from Gamalytic, a platform that aggregates game performance data by pulling directly from Steam's public API and other publicly available Steam data. Gamalytic compiles this information into structured, accessible datasets well-suited for analysis.

It is important to note that not all columns in the dataset represent raw, directly observed values. Several fields, most notably revenue, are estimated by Gamalytic using a proprietary algorithm rather than reported directly by Steam or game developers. Gamalytic's revenue estimates are derived from a combination of factors including Steam review counts, peak concurrent player data from Steam Charts, median review-to-sales multipliers, and publicly available sales disclosures used to calibrate results. This approach reflects a well-known industry method sometimes referred to as the review-to-sales ratio, where historical patterns between user reviews and total sales are applied to estimate units sold, which is then multiplied by price to approximate gross revenue.

Gamalytic has self-assessed the accuracy of these estimates, reporting that 77% of individual game estimates fall within a 30% margin of error, and 98% fall within a 50% margin of error, while aggregate-level estimates reach approximately 99% accuracy. Smaller games are noted to have less accurate estimates than the larger ones due to smaller sample size. Given these limitations, the revenue figures used in this analysis should be interpreted as reasonable approximations rather than exact values, and any findings involving revenue should be contextualized accordingly.

2.2 Dataset Overview

Two datasets were used in this analysis, both originating from Gamalytic and sharing the same base population of 10,638 Steam game titles. Such games were already partially filtered, with all games in the dataset being established games with 500+ reviews, revenue greater than 0, and base game price greater than 0 (this got rid of free-to-play games which would severely skew revenue data).

The primary dataset (dataset.csv) contains 10 variables covering core game attributes: name, release date, copies sold, price, revenue, average playtime, review score, publisher class, publisher, and developer. This leaner dataset served as the foundation for statistical testing, regression modeling, and revenue group comparisons.

The expanded dataset (games-list-dataset-1781018552421.csv) contains 12 variables across the same 10,638 titles, adding review count and wishlists. The Random Forest model used this expanded export because it included additional predictive variables needed for machine learning, especially review count and wishlists. After excluding Hobbyist titles ($n = 58$) and removing incomplete model rows, the Random Forest used 10,572 complete observations, split into 8,487 training rows and 2,085 testing rows.

2.3 Variable Descriptions

Variable	Type	Description	Use
name	Text	Title of the Steam game	Identifier variable
releaseDate	Date	Date the game was released on Steam	Used for temporal context and filtering
copiesSold	Numeric	Estimated total units sold	Supporting sales volume variable
price	Numeric	Game retail price in USD	Evaluates pricing effects on revenue
revenue	Numeric	Estimated gross revenue (USD)	Primary outcome variable measuring financial performance
averagePlaytime	Numeric	Mean hours played per player	Engagement variable capturing player retention
reviewScore	Numeric	Aggregate player review score (0–100 scale)	Measures player reception and game quality
publisherClass	Categorical	Publisher class: AAA , Indie , Hobbyist	Primary grouping variable for publisher-tier comparisons
publisher	Text	Name of the game's publishing entity	Used for publisher-level identification

developer	Text	Name of the game's development studio	Used for developer-level identification
Review count	Numeric	Total volume of player reviews submitted	Random Forest predictor measuring scale of player response
Wishlists	Numeric	Number of Steam wishlists	Random Forest predictor measuring pre-purchase demand

2.4 Scope & Limitations

This analysis uses Gamalytic Steam analytics data, including estimated revenue, sales, pricing, reviews, wishlists, publisher class, and playtime fields. Gamalytic compiles Steam-related performance data from Steam's public API and other publicly available Steam data sources. Results should be interpreted as market-performance estimates, not official Steam or developer-reported financial records.

A key limitation is that some fields, especially revenue, are estimated rather than directly observed. Gamalytic estimates revenue using a proprietary method informed by signals such as Steam review counts, player activity data, review-to-sales ratios, and publicly available sales disclosures. Gamalytic has reported that most individual estimates fall within broad error margins, while aggregate estimates are more accurate. Smaller or lower-activity games may have less reliable estimates because there is less public signal available.

Review tier analysis is limited by uneven group sizes. Most games fall into the positive review tiers, while the negative review tiers contain very few games, so comparisons involving Mostly Negative and Overwhelmingly Negative titles should be interpreted carefully. Also, because review tiers group a continuous review score into categories, they help interpretation but simplify variation within each score range.

The final analysis focuses on AAA, AA, and Indie games, with Hobbyist titles excluded for consistency across the main comparison and modeling work. The dataset supports analysis across publisher class, pricing, review performance, engagement, and market demand, but it does not include outside factors such as marketing budget, franchise strength, genre popularity, streamer exposure, launch timing, platform visibility, discounting history, or broader market trends.

The regression model was useful for estimating direct relationships between selected predictors and log revenue while controlling for other variables. However, regression assumes a structured linear relationship and may not fully capture nonlinear patterns or interactions between price, reviews, engagement, and publisher class. If outliers were removed during regression, that improves stability for typical games but makes the model less representative of extreme breakout titles.

The Random Forest model provided stronger predictive performance by capturing nonlinear patterns and interactions across variables. It used 10,572 complete AAA, AA, and Indie observations and achieved R-squared = 0.906, RMSE = 0.242, and MAE = 0.189 on the test set. However, the model is predictive rather than causal. Because Gamalytic revenue estimates may partially rely on review activity, the strong importance of review_count should be interpreted carefully: it reflects market attention, but may also partly reflect how the estimated revenue variable was constructed.

To reduce any data leakage, direct revenue-linked fields such as revenue, copies sold, log copies sold, total revenue, player totals, and M1/Y1 revenue or sales fields were excluded from predictive modeling. Price and review tiers were grouped using the unified project definitions, which improves interpretation but simplifies variation within each tier. Overall, the findings are best understood as a structured view of estimated Steam revenue patterns, not a complete causal explanation of game success.

3. Data Preparation & Cleaning

3.1 Missing Values

The raw dataset was loaded into R using `read.csv()`, with both empty strings and "NA" values treated as missing using the `na.strings = c("", "NA")` argument. The initial dataset contained 10,638 rows and 10 columns. Using `colSums(is.na(data))`, missing values were found in two columns: publisher had 42 NAs and developer had 15 NAs. All numeric variables had zero missing values. To handle this, `na.omit()` was used to remove any rows containing missing data. Since the number of affected rows was small relative to the full dataset, this approach had minimal impact on the overall analysis.

3.2 Outlier Handling

All rows where publisherClass was labeled "Hobbyist" were removed using `data[data$publisherClass != "Hobbyist", ]`, since hobbyist titles are not comparable to commercial releases and would have skewed analysis results.

3.3 Data Transformation

Before modeling, the skewness of all five numeric variables was checked using the moments library, and each variable was transformed based on the magnitude and direction of its skew. `copiesSold` and `revenue` were both heavily right-skewed, so a `log10` transformation was applied to each. `averagePlaytime` was also heavily skewed, but since some values were zero, `log10(averagePlaytime + 1)` was used to avoid taking the log of zero. `price` had moderate right skew, so a square root transformation was applied. `reviewScore` was moderately left-skewed, so a reflect-and-square-root approach was used, `sqrt(k - reviewScore)` where `k = max(reviewScore) + 1`, to flip and correct the distribution. After transforming, all five variables had skewness values under 0.85, indicating that their distributions were much closer to normal due to effective transformations. The `releaseDate` column was also converted from a raw string into a proper date format using `as.POSIXct()` followed by `as.Date()`.

3.4 Variables Excluded

The "Hobbyist" category from publisherClass was excluded because hobbyist developers operate very differently from commercial ones, and including them would make it harder to draw meaningful comparisons across publisher types.

3.4 Appended Dataset for Random Forest

The only difference compared to the first dataset is that the following columns were added:

- Review count — the total number of player reviews submitted for a game, serving as a proxy for overall sales volume and community engagement
- Wishlists — the number of times a game was wishlisted on Steam prior to or after release, serving as a measure of pre-purchase consumer demand

These two variables were included exclusively in the Random Forest model as additional predictors, as they are not part of the core dataset used for statistical testing. Eight observations with missing Wishlists values were removed prior to model training. All other cleaning steps,

Hobbyist exclusion and removal of incomplete rows, were applied identically to this expanded dataset, yielding 10,572 complete observations used for modeling.

4. Dimension Analyses

4.1 Dimension 1: Publisher Class and Revenue Performance: AAA vs. AA vs. Indie

4.1.1 Overview & Business Question

Dimension 1 examines how publisher classification relates to revenue performance in the Steam marketplace. For this section, games were grouped into three major publisher classes: AAA, AA, and Indie. Hobbyist titles were excluded from this comparison because the group was much smaller than the other publisher categories, making the AAA, AA, and Indie comparison cleaner and more reliable.

This comparison matters because games do not enter the Steam marketplace with the same level of funding, visibility, marketing support, brand recognition, pricing power, or audience expectations. AAA games often benefit from larger budgets, stronger promotional support, and greater market awareness before launch. Indie games usually compete through lower prices, niche appeal, player reviews, creativity, and community support. AA games sit between these two groups, often having more resources than Indie titles but less reach than major AAA releases.

The main business question for this dimension is:

How does publisher classification relate to Steam revenue performance?

This section examines whether AAA, AA, and Indie games show meaningful differences in revenue, price, reviews, player activity, and overall market share. The goal is to better understand whether publisher class helps explain which games tend to generate stronger financial outcomes on Steam.

4.1.2 Descriptive Statistics

The filtered Dimension 1 dataset included 10,580 games across the three publisher classes. Indie games made up the largest share of the dataset by count, with 7,440 titles, or 70.3% of all games analyzed. AA games accounted for 2,112 titles, or 20.0%, while AAA games made up 1,028 titles, or 9.7% of the dataset. This shows that Indie games dominate the dataset in terms of volume.

However, the revenue results tell a different story. Even though AAA games represented less than 10% of the analyzed titles, they accounted for the largest share of total revenue at 44.3%, or about \$27.2 billion. AA games followed with 36.2% of total revenue, or about \$22.2 billion. Indie games, despite representing most of the dataset by count, accounted for 19.5% of total revenue, or about \$12.0 billion.

Median revenue also showed a clear difference between publisher classes. AAA games had the highest median revenue at about \$5.1 million, followed by AA games at about \$1.4 million. Indie games had a much lower median revenue at about \$407.2 thousand. This means the typical AAA game in the dataset earned about 12.6 times more revenue than the typical Indie game.

Price also differed by publisher class. AAA games had the highest median price at \$24.99, followed by AA games at \$19.99 and Indie games at \$14.99. The price tier breakdown showed that 21.6% of AAA games were priced at \$40 or higher, compared with 9.6% of AA

games and 1.3% of Indie games. Indie games were mostly concentrated under \$20, with 82.7% of Indie titles falling into that lower price tier.

4.1.3 Analysis & Findings

Publisher class meaningfully affects revenue performance, but not through game count alone. Indie games make up most of the dataset at 70.3% of titles, while AAA games account for the largest revenue share at 44.3%. AA games contribute 36.2%, and Indie games contribute 19.5%. This shows that Steam's title volume is driven by Indie games, while revenue is more concentrated among AAA and AA publishers.

The statistical tests support this pattern. Because the variance test showed unequal group variances, Welch ANOVA was used as the main test. Welch ANOVA found a significant difference in mean log revenue across publisher classes, $F = 1095.061$, $p < 0.001$. Pairwise Welch tests showed that all three groups differ significantly from each other: AAA vs AA, AAA vs Indie, and AA vs Indie.

The adjusted regression also supports the publisher-class effect. After controlling for price, review score, and average playtime, both AA and Indie games had significantly lower log revenue than AAA games. This means publisher class remains meaningful even after accounting for other performance factors.

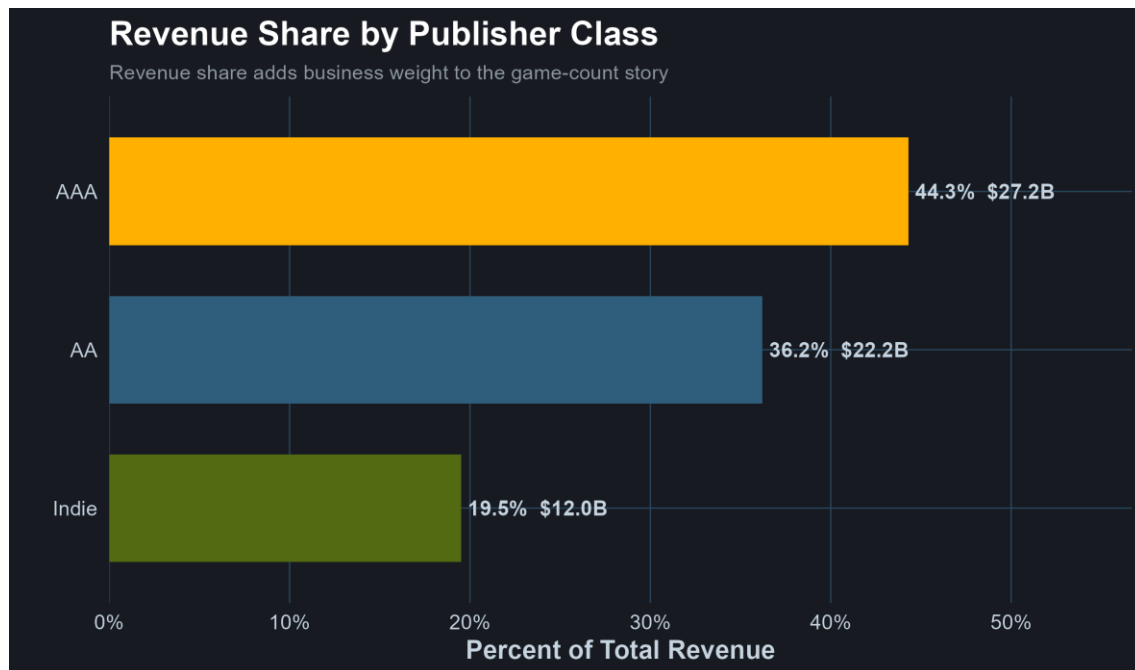
The correlation matrix adds context. Sqrt price had the strongest relationship with log revenue, $r = 0.621$, followed by log average playtime, $r = 0.502$. Log review score had a much weaker relationship, $r = 0.107$. This suggests that pricing power and player engagement are stronger revenue signals than review score alone.

Overall, Dimension 1 shows that publisher class reflects meaningful differences in market position and revenue outcomes. Indie games lead in title count and median review score, but AAA games lead in revenue share, median revenue, price, and playtime. AA games sit between AAA and Indie, with revenue performance stronger than Indie but still significantly below AAA.

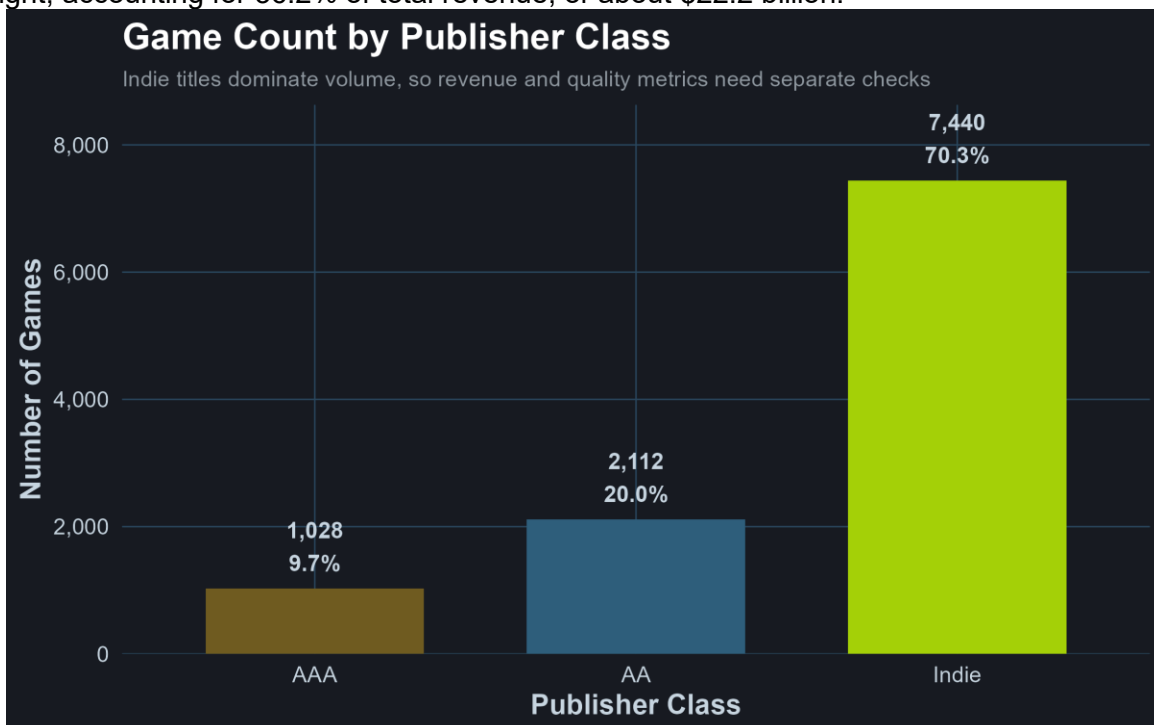
4.1.4 Visualizations

Dimension 1 examines how publisher class relates to revenue performance across AAA, AA, and Indie games on Steam. The goal is to determine whether publisher classification helps explain differences in revenue, pricing, engagement, and overall market share.

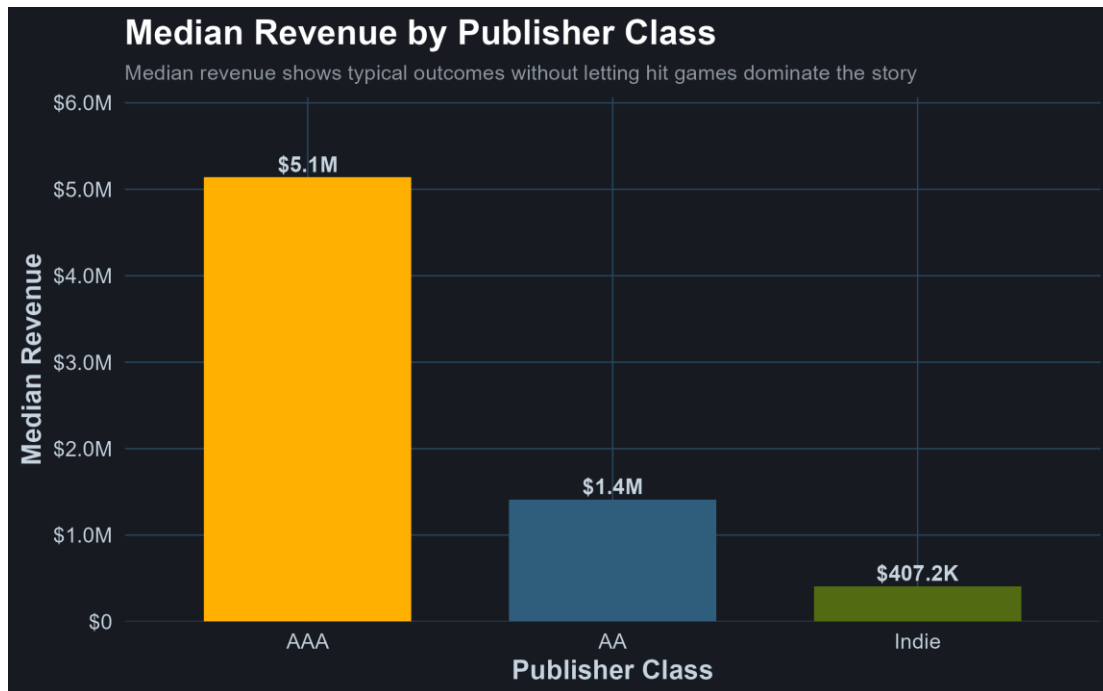
The main finding is that Indie games dominate by title count, but AAA and AA games carry most of the financial weight. This suggests that publisher class reflects more than a label; it reflects differences in market position, pricing power, visibility, and player engagement.



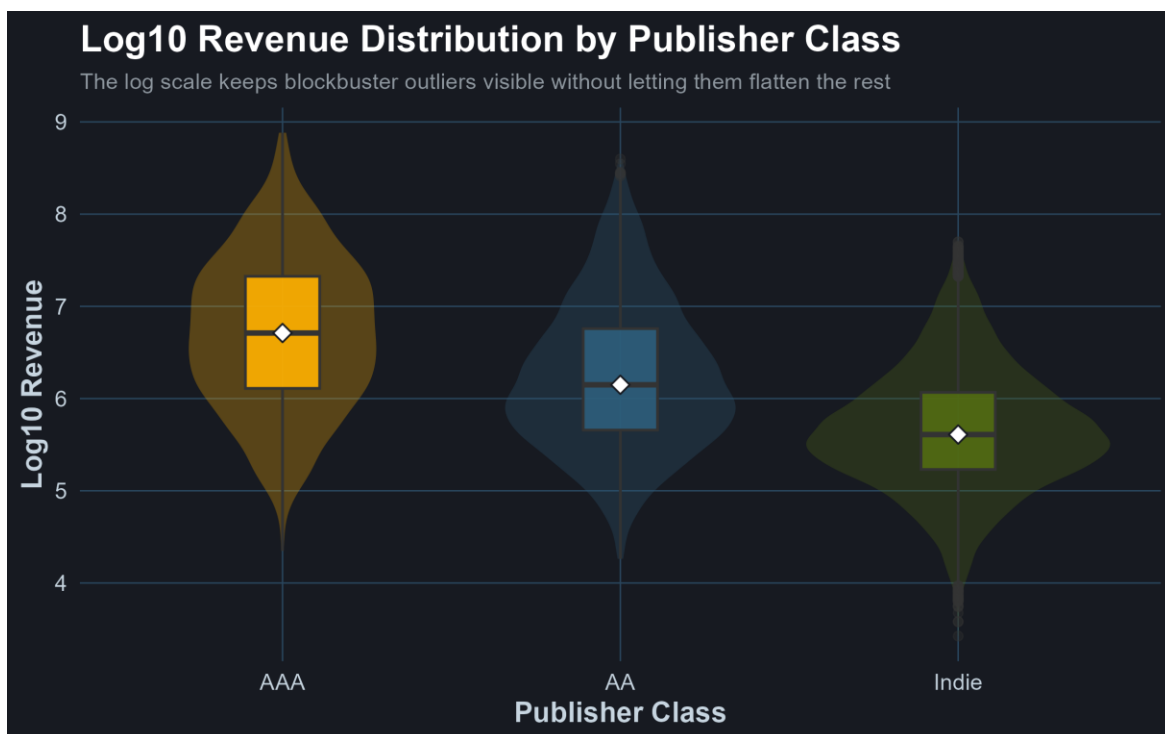
The revenue share visualization highlights the structural gap between release volume and financial capture. Indie games make up 70.3% of analyzed titles, but only 19.5% of total revenue, or about \$12.0 billion. In contrast, AAA games represent only 9.7% of titles but lead total revenue share at 44.3%, or about \$27.2 billion. AA games also carry strong financial weight, accounting for 36.2% of total revenue, or about \$22.2 billion.



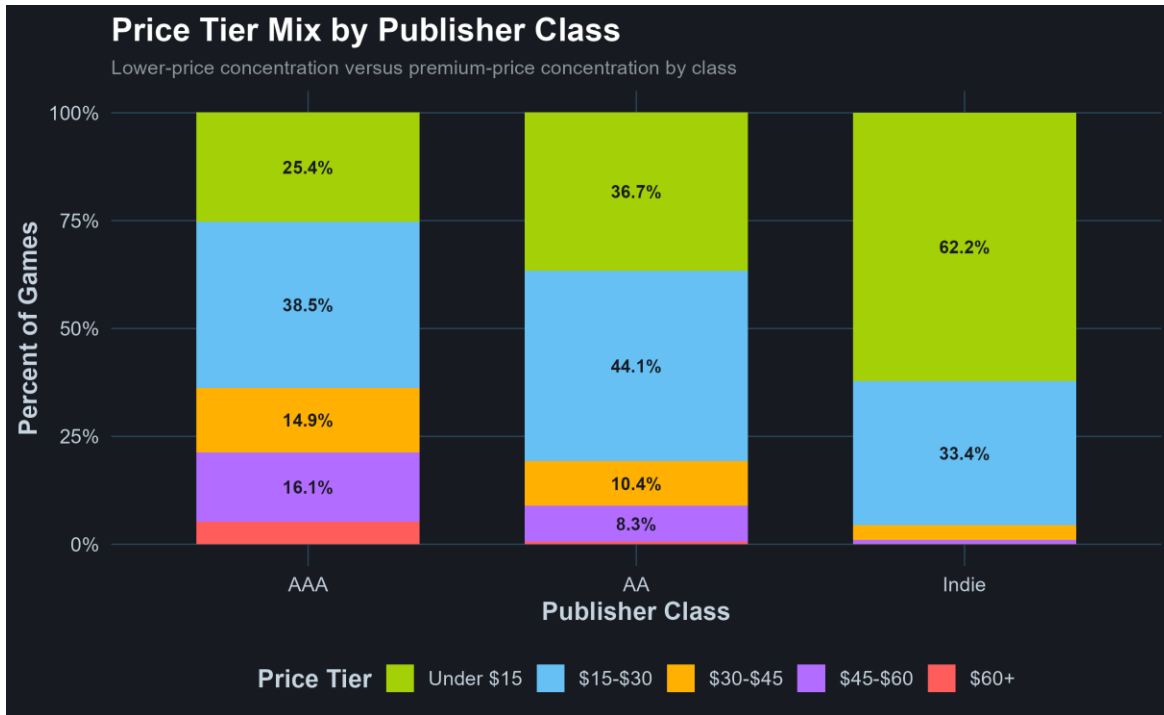
This visual shows that Indie games make up the largest share of the dataset, with 7,440 titles, or 70.3% of analyzed games. AA games account for 20.0%, while AAA games make up only 9.7%, showing that Steam's catalog volume is heavily driven by Indie titles.



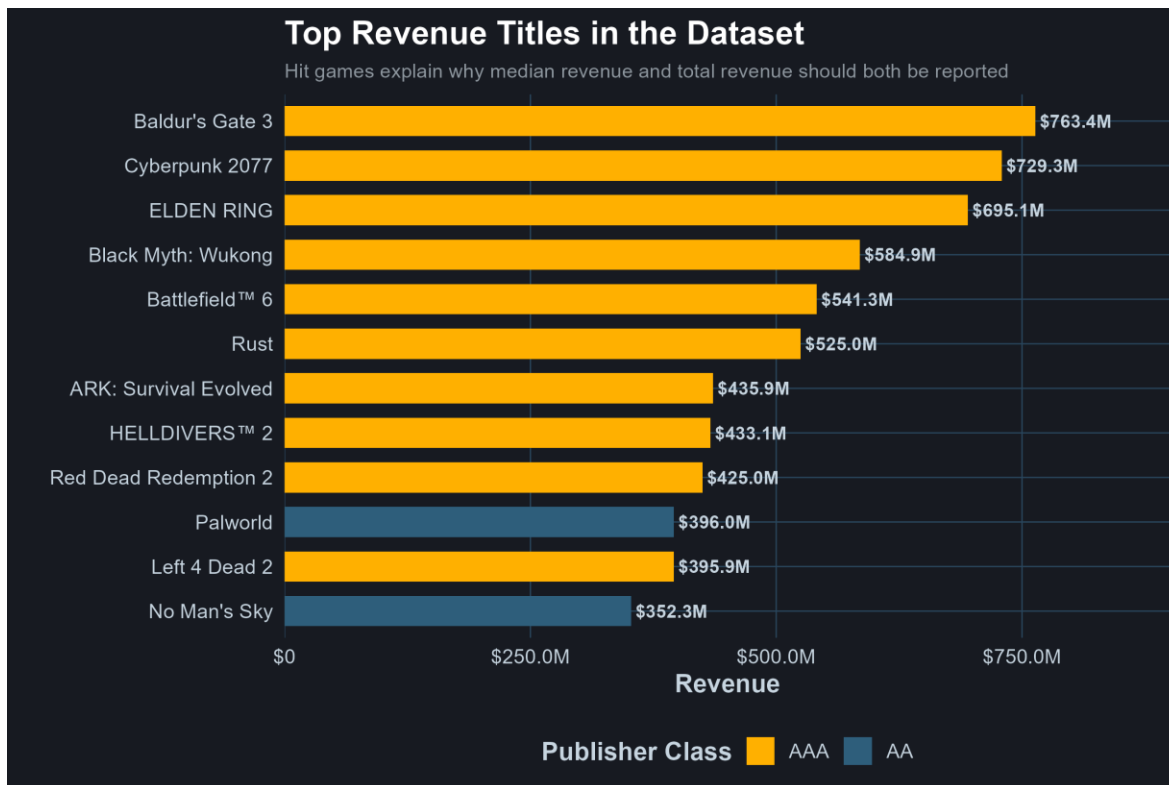
Median revenue shows the typical revenue outcome within each publisher class without letting blockbuster titles dominate the result. AAA games had the highest median revenue at about \$5.1 million, followed by AA at \$1.4 million and Indie at \$407.2 thousand.



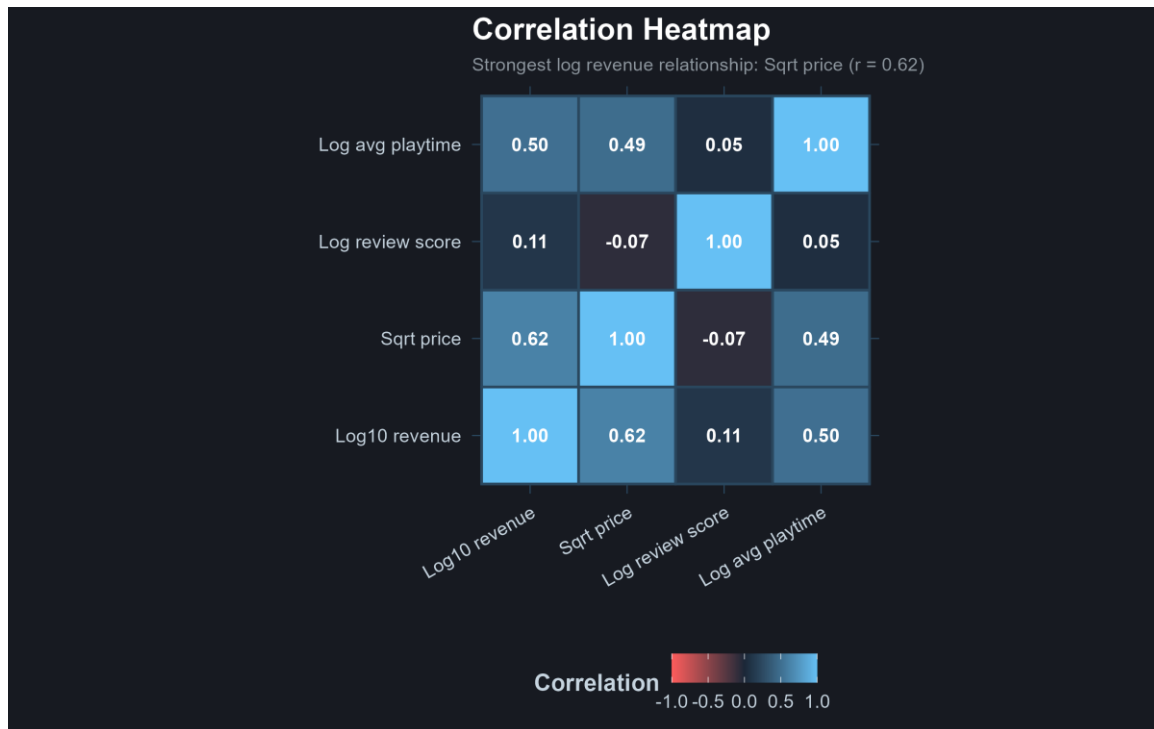
The log revenue distribution shows that AAA games generally sit higher on the revenue scale, with AA in the middle and Indie lower overall. Using log revenue helps keep blockbuster games visible while still allowing the broader revenue pattern to be compared fairly.



The Random Forest model used 10,572 complete AAA, AA, and Indie observations after excluding Hobbyist titles. Using unified price and review tier definitions, the model achieved an R-squared of 0.906, RMSE of 0.242, and MAE of 0.189 on the test set.



The top revenue titles show why total revenue and median revenue both matter. A small number of hit games generate hundreds of millions of dollars, which can heavily influence total revenue and make median revenue useful for understanding typical performance.



The correlation heatmap shows that sqrt price had the strongest relationship with log revenue at $r = 0.62$. Log average playtime also had a moderate relationship with log revenue at $r = 0.50$, while log review score was much weaker at $r = 0.11$.

Descriptive Statistics - Overall Numeric Variables							
Variable	N	Missing	Mean	SD	Min	Median	Max
Revenue	10,580	0	5,808,061.000	26,590,055.000	2,657	601,502	763,418,784
Log Revenue	10,580	0	5.878	0.801	3.424	5.779	8.883
Total Revenue	664	9,916	29,147,106.000	71,722,578.000	379,124	8,247,011.000	785,671,547
Price	10,580	0	18.114	12.901	1.450	14.990	99.990
Sqrt Price	10,580	0	4.009	1.428	1.204	3.872	9.999
Copieessold	10,580	0	462,228.500	1,803,594.000	1,501	75,209	67,437,582
Players Total	10,580	0	584,335.900	2,145,312.000	5,001	105,191.500	80,439,417
Avgplaytime	10,580	0	13.121	19.507	0	7.232	472.809
Log Avgplaytime	10,580	0	0.971	0.359	0	0.915	2.676
Reviewscore	10,580	0	82.862	12.402	10	86	100
Review Count	10,580	0	11,520.220	54,577.330	500	1,670	1,527,153
Wishlists	10,572	8	203,842.900	379,259.000	108	86,798	6,342,598
M1 Revenue	8,618	1,962	1,881,805.000	11,313,165.000	93	180,556.500	413,232,824
M1 Sales	8,615	1,965	73,300.500	337,958.700	15	14,776	12,867,165
Y1 Revenue	8,392	2,188	3,055,099.000	14,993,915.000	427	340,278.500	546,496,151
Y1 Sales	8,388	2,192	148,780.400	563,562.100	98	33,276	17,790,068

Revenue, sales, review, playtime, and wishlist variables are summarized after filtering the dataset to AAA, AA, and Indi

Correlation Matrix - Transformed Numeric Variables

Variable	Log10 revenue	Sqrt price	Log review score	Log avg playtime
Log10 revenue	1	0.621	0.107	0.502
Sqrt price	0.621	1	-0.071	0.485
Log review score	0.107	-0.071	1	0.050
Log avg playtime	0.502	0.485	0.050	1

Correlation values summarize linear relationships among transformed numeric variables used in the pu

4.2 Dimension 2: High vs. Low Price Tier

4.2.1 Overview & Business Question

This dimension examines how a game's price tier relates to its net revenue among Steam games in the expanded dataset. Games were categorized into three pricing tiers derived from the original price column: Under \$20, \$20–\$40, and \$40+. The central business question(s) this dimension addresses is, *among Steam games, which price tier is associated with the highest net revenue, and are those differences statistically significant?*

4.2.2 Descriptive Statistics

All descriptive statistics and statistical testing were conducted on log-transformed revenue unless otherwise noted. Raw revenue figures are reported alongside log-transformed values for interpretability. There is also a significant imbalance across the tiers, particularly the \$40+ price tier, and should be considered when interpreting these results. See *Section 2.4* for a full discussion of dataset scope and limitations.

As seen in Table 4.2.1, the Under \$20 price tier dominates the dataset accounting for approximately 75% of all games. Despite this, Under \$20 has the lowest mean log revenue (5.67), suggesting that volume alone does not compensate for lower price points in the budget segment. The \$40+ tier, while representing only 4.9% of games, has the highest mean log revenue (6.93), reinforcing that premium pricing is associated with substantially higher revenue outcomes.

As seen in Figure 4.2.1, the bar chart illustrates the average log revenue by price tier. It shows a clear stepwise increase in mean log revenue from Under \$20 (5.67) through \$20–\$40 (6.40) up to \$40+ (6.93). The visual gap between the \$40+ bar and the other tiers is the largest, reinforcing that premium pricing is associated with substantially higher revenue.

As seen in Figure 4.2.3, the bar chart illustrates the significant imbalance in game counts across the tiers. The Under \$20 bar dominates at 7,985 games, while \$20–\$40 accounts for 2,132 games and \$40+ for 521 games. This distribution reinforces that while the budget tier dominates in volume, it is the least financially rewarding among the analyzed Steam games.

4.2.3 Analysis & Findings

As seen in Table 4.2.2, a correlation matrix was computed on all transformed variables to assess the relationship between price and revenue. The correlation between `sqrt_price` and `log_revenue` was $r = 0.62$, which indicates a strong positive relationship between price and revenue as continuous variables, a notable improvement from the original dataset where this correlation was only $r = 0.30$. `log_avgPlaytime` also showed a strong correlation with `log_revenue` ($r = 0.50$), suggesting that player engagement remains an important predictor of revenue alongside price.

As seen in Table 4.2.3, a one-way ANOVA was conducted to test whether price tier has a statistically significant effect on log-transformed revenue. The results indicated $F(2, 10635) = 1481$, $p < 2e-16$, confirming that price tier meaningfully differentiates revenue outcomes among Steam games in this dataset.

As seen in Table 4.2.4, a Tukey post-hoc test was applied to identify which specific tier pairs differ significantly from one another, controlling for family-wise Type I error. With three price tiers, there are three pairwise comparisons. All three comparisons were statistically significant ($p \approx 0$), confirming that each price tier generates meaningfully different revenue outcomes.

As seen in Figure 4.2.2, the boxplot illustrates the log revenue distribution by price tier. The \$40+ tier shows the highest median log revenue and the widest interquartile range (IQR), reflecting both high central tendency and considerable spread among premium titles. The \$20–\$40 tier shows a moderately wide IQR with some high-revenue outliers, suggesting a small number of very successful mid-range titles. The Under \$20 tier shows the tightest IQR but the most outliers above the upper whisker, indicating that while most budget games cluster at lower revenue levels, a handful of breakout titles reach significantly higher earnings.

As seen in Figure 4.2.4, the scatterplot illustrates a scatterplot of original price against log revenue, color-coded by tier. The dense cluster of blue points (Under \$20) concentrated at the left of the x-axis confirms the volume dominance of budget titles. The sparse yellow points (\$40+) spread across higher price values and tend to occupy the upper range of the y-axis. The scatterplot also visually confirms the strong positive relationship between price and log revenue ($r = 0.62$), as the upward trend across tiers is clearly visible.

4.2.4 Visualizations

Table 4.2.1: Descriptive Statistics

Price Tier	Count	% of Dataset	Mean Revenue	Median Revenue	Mean Log Revenue
Under \$20	7,920	74.9%	\$2,548,808	\$393,644	5.67
\$20-\$40	2,127	20.1%	\$10,701,204	\$2,070,652	6.40
\$40+	521	4.9%	\$35,297,188	\$7,354,687	6.93

Table 4.2.2: Correlation Matrix

	log_revenue	log_copiesSold	sqrt_price	log_avgPlaytime	sqrt_reviewScore
log_revenue	1.0000	0.8758	0.6216	0.5017	-0.1005
log_copiesSold	0.8758	1.0000	0.2672	0.3392	-0.1463
sqrt_price	0.6216	0.2672	1.0000	0.4851	0.0977
log_avgPlaytime	0.5017	0.3392	0.4851	1.0000	-0.0385
sqrt_reviewScore	-0.1005	-0.1463	0.0977	-0.0385	1.0000

Table 4.2.3: One-Way ANOVA

Source	df	Sum Sq	Mean Sq	F value	p value
Price Tier	2	1485	742.6	1482	<2e-16 ***
Residuals	10565	5294	0.5	-	-

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

Table 4.2.4: Post-Hoc Tukey Test

Comparison	Difference	Significant?
\$20–\$40 vs Under \$20	0.726	Yes (p ≈ 0)
\$40+ vs Under \$20	1.256	Yes (p ≈ 0)
\$40+ vs \$20–\$40	0.530	Yes (p ≈ 0)

Figure 4.2.1: Bar chart (x = Price Tier, y = Mean Log10 Revenue)

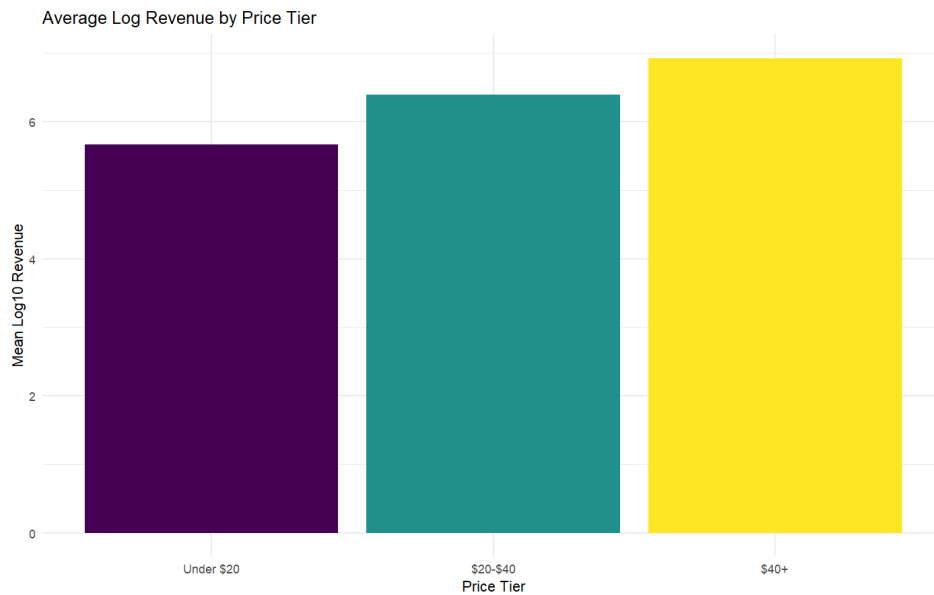


Figure 4.2.2: Box plot (x = Price Tier, y = Log10 Revenue)

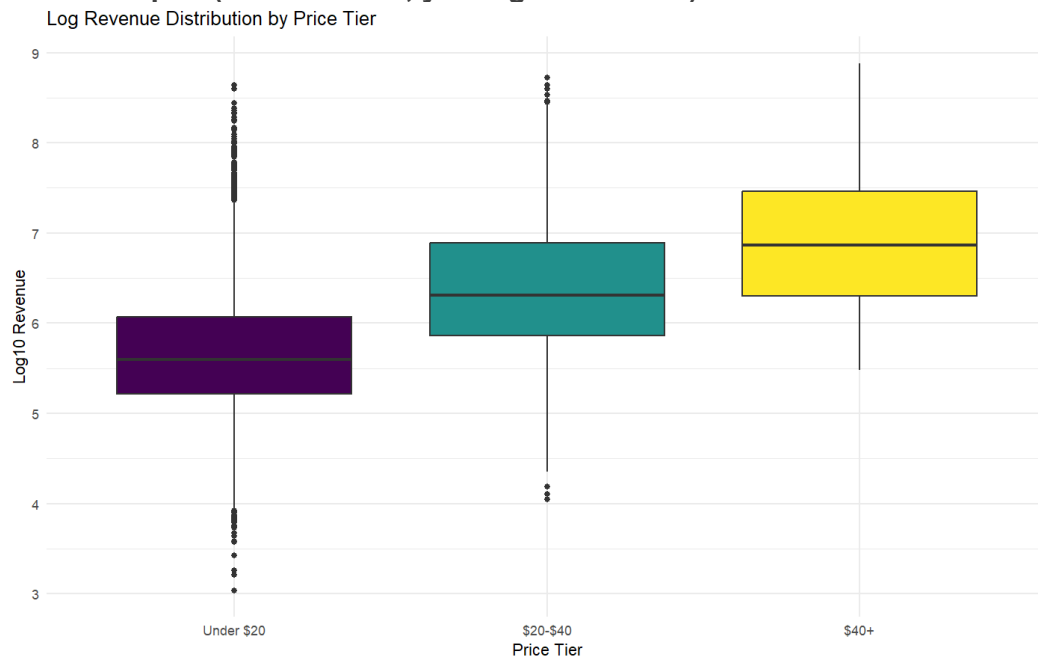


Figure 4.2.3: Bar chart (x = Price Tier, y = Count)

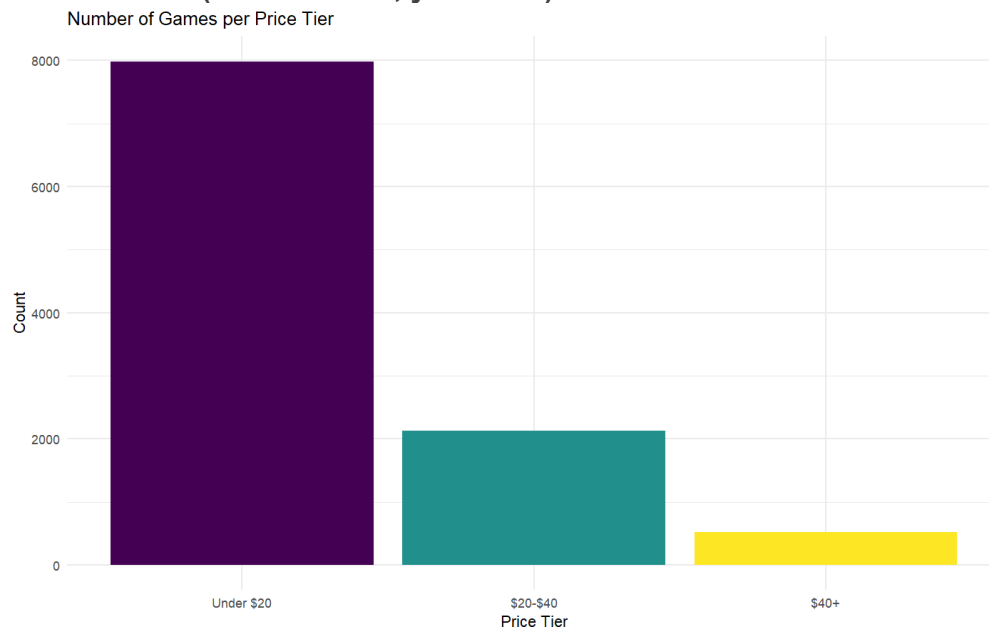
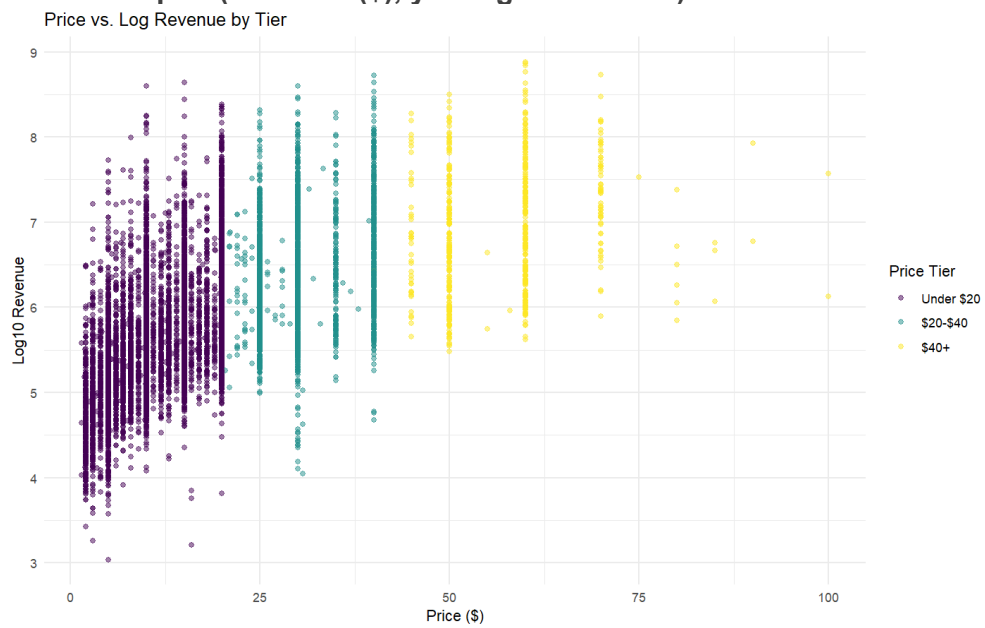


Figure 4.2.4: Scatterplot (x = Price (\$), y = Log10 Revenue)



4.3 Dimension 3: High vs. Low Review Score

4.3.1 Overview & Business Question

This dimension examines how a game's review score relates to its net revenue among Steam games in the dataset. Games were categorized into six review score tiers using Steam's own review score tier criteria and are derived from the original review score column: Overwhelmingly Positive ($\geq 95\%$), Very Positive (80–94%), Mostly Positive (70–79%), Mixed (40–69%), Mostly Negative (20–39%), and Overwhelmingly Negative ($< 20\%$). The central business question this dimension addresses is: Among Steam games, which review score tier is associated with the highest net revenue, and are those differences statistically significant? More broadly, this analysis investigates whether critical reception, as measured by the percentage of positive user reviews, translates to meaningful commercial success on Steam.

4.3.2 Descriptive Statistics

As shown in Table 4.3.1, the dataset is heavily concentrated in the positive review tiers, with Very Positive (80–94%) being the largest group at 4,191 games (39.7% of the dataset), followed by Mostly Positive at 3,466 games (32.8%). The two negative tiers are extremely small, Mostly Negative contains only 74 games (0.7%) and Overwhelmingly Negative just 6 games (0.1%), meaning that the findings for those groups carry limited statistical reliability.

Mean revenue rises with review tier in a broadly monotonic pattern: Overwhelmingly Positive games lead at \$9.50M mean revenue, while Mostly Negative games trail at \$2.44M. However, mean figures are heavily distorted by outliers. The Overwhelmingly Positive tier's \$9.50M mean against a median of only \$649,204 reveals that a small number of blockbuster titles are pulling the average upward, not a broad revenue lift across all highly-rated games. The Figure 4.3.2 box plot reinforces this: the Overwhelmingly Positive and Very Positive tiers have the widest revenue distributions and highest concentration of extreme outliers, while Mixed and Mostly Negative tiers show tighter, lower distributions.

On the log scale, mean log revenues are closely clustered: 5.94 (Overwhelmingly Positive), 5.93 (Very Positive), 5.85 (Mostly Positive), 5.72 (Mixed), and 5.48 (Mostly Negative), as confirmed in Table 4.3.1. These differences, while statistically detectable, are modest in practical terms as the gap between Mixed and Very Positive on a log scale translates to a relatively small real-world revenue advantage.

4.3.3 Analysis & Findings

The correlation matrix (Table 4.3.2) reveals that review score has a near-zero linear relationship with log revenue ($r = -0.1005$), and is similarly negligible against average playtime ($r = -0.0385$) and price ($r = 0.0977$). This directly challenges the intuitive assumption that earning better reviews leads to higher earnings. In contrast, log copies sold ($r = 0.8758$) and sqrt price ($r = 0.6216$) are far stronger revenue signals, suggesting that market demand and pricing strategy outweigh review quality as revenue drivers.

The one-way ANOVA (Table 4.3.3) confirms that review score tier does have a statistically significant relationship with mean log revenue overall ($F = 20.63$, $df = 5$, $p < 2e-16$). However, statistical significance at this scale does not imply large practical differences. The post-hoc

Tukey test (Table 4.3.4) reveals that most meaningful separations occur when comparing the top tiers against Mixed and below. For example, Mixed vs. Overwhelmingly Positive (diff = -0.2244 , $p < 0.0001$) and Mostly Negative vs. Overwhelmingly Positive (diff = -0.4590 , $p < 0.0001$) are both significant. Crucially, Very Positive vs. Overwhelmingly Positive is not significant (diff = -0.0096 , $p = 0.9986$), meaning the revenue difference between the two highest tiers is essentially negligible.

The scatterplot in Figure 4.3.4 (Review Score vs. Log Revenue by Tier) further illustrates the weak relationship: points are dispersed widely across all review score levels with no clear upward trend, and high-revenue games appear across the full review score range. The Figure 4.3.3 bar chart shows the distributional imbalance. Very Positive and Mostly Positive together account for over 72% of all games, while the negative tiers are near-absent, which limits the analytical power of any comparisons involving those groups. Taken together, the evidence indicates that while reaching Overwhelmingly Positive status ($\geq 95\%$) is associated with higher average revenue, the effect is driven by outliers, and moving from Mixed to Very Positive yields only modest revenue gains, meaning that review scores alone are a weak and unreliable lever for financial success on Steam.

4.3.4 Visualizations

Table 4.3.1: Descriptive Statistics

Price Tier	Count	% of Dataset	Mean Revenue	Median Revenue	Mean Log Revenue
Overwhelmingly Positive ($\geq 95\%$)	1,543	14.6%	\$9.50M	\$649,204	5.94
Very Positive (80–94%)	4,191	39.7%	\$6.19M	\$671,959	5.93
Mostly Positive (70–79%)	3,466	32.8%	\$4.51M	\$584,481	5.85
Mixed (40–69%)	1,288	12.2%	\$3.82M	\$448,736	5.72
Mostly Negative (20–39%)	74	0.7%	\$2.44M	\$335,443	5.48
Overwhelmingly Negative ($< 20\%$)	6	0.1%	\$836,671	\$740,068	5.71

Table 4.3.2: Correlation Matrix

	log_revenue	sqrt_reviewScore	log_avgPlaytime	log_copiesSold	sqrt_price
--	-------------	------------------	-----------------	----------------	------------

log_revenue	1.0000	-0.1005	0.5017	0.8758	0.6216
sqrt_reviewScore	-0.1005	1.0000	-0.0385	-0.1463	0.0977
log_avgPlaytime	0.5017	-0.0385	1.0000	0.3392	0.4851
log_copiesSold	0.8758	-0.1463	0.3392	1.0000	0.2672
sqrt_price	0.6216	0.0977	0.4851	0.2672	1.0000

Table 4.3.3: One-Way ANOVA

Source	df	Sum Sq	Mean Sq	F value	p value
Review Score Tier	5	66	13.111	20.63	<2e-16 ***
Residuals	10562	6714	0.636	-	-

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

Table 4.3.4: Post-Hoc Tukey Test

Comparison	Difference	Significant?
Very Positive vs Ovwh. Positive	-0.0096	No (p=0.9986)
Mostly Positive vs Ovwh. Positive	-0.0879	Yes (p=0.0043)
Mixed vs Ovwh. Positive	-0.2244	Yes (p is < 0.0001)
Mostly Negative vs Ovwh. Positive	-0.4590	Yes (p is < 0.0001)
Ovwh. Negative vs Ovwh. Positive	-0.2323	No (p=0.9805)
Mostly Positive vs Very Positive	-0.0783	Yes (p=0.0003)
Mixed vs Very Positive	-0.2148	Yes (p is < 0.0001)
Mostly Negative vs Very Positive	-0.4494	Yes (p is < 0.0001)
Ovwh. Negative vs Very Positive	-0.2227	No (p=0.9838)
Mixed vs Mostly Positive	-0.1366	Yes (p is < 0.0001 *)
Mostly Negative vs Mostly Positive	-0.3711	Yes (p=0.0011)
Ovwh. Negative vs Mostly Positive	-0.1445	No (p=0.9978)
Mostly Negative vs Mixed	-0.2345	No (p=0.1360)
Ovwh. Negative vs Mixed	-0.0079	No (p=1.0000)
Ovwh. Negative vs Mostly Negative	0.2266	No (p=0.9852)

Figure 4.3.1: Bar chart (x = Price Tier, y = Mean Log10 Revenue)

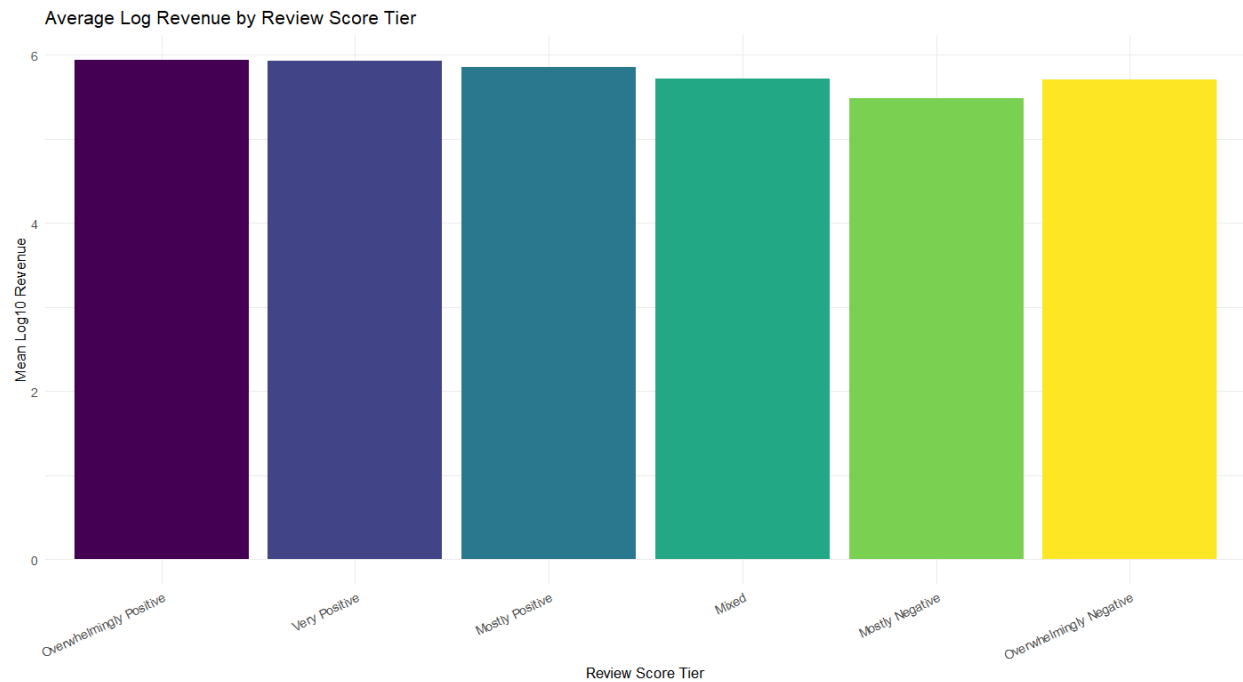


Figure 4.3.2: Box plot (x = Review Score Tier, y = Log10 Revenue)

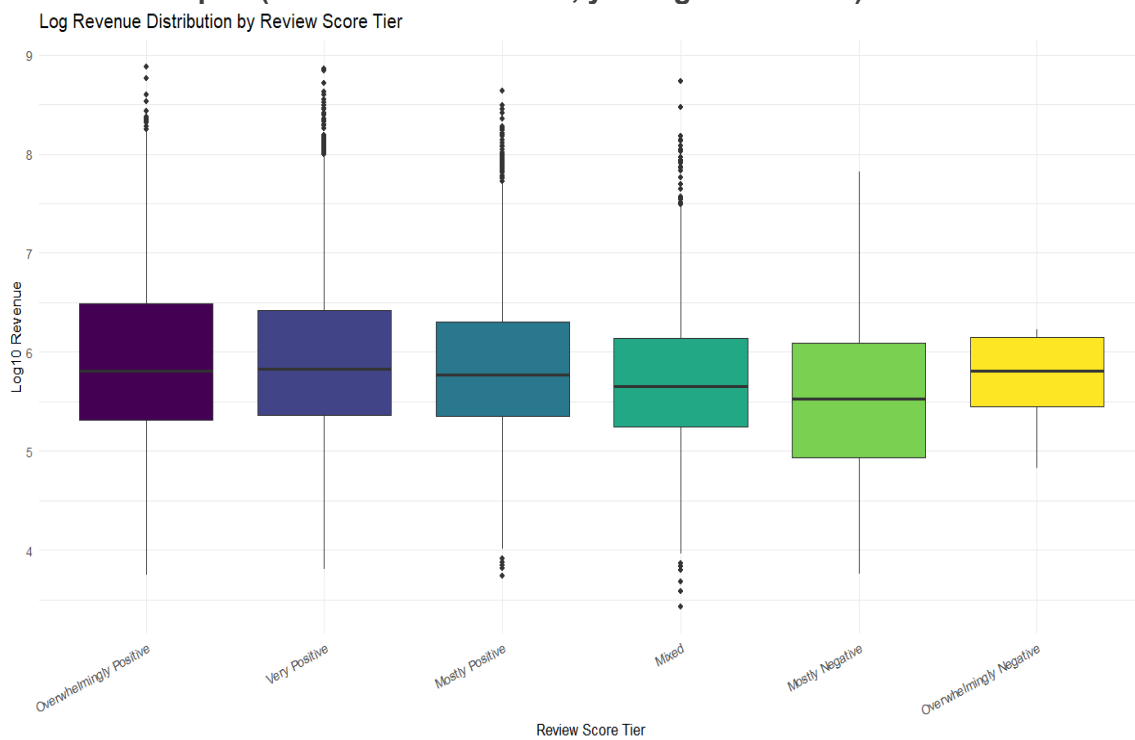


Figure 4.3.3: Bar chart (x = Review Score Tier, y = Count)

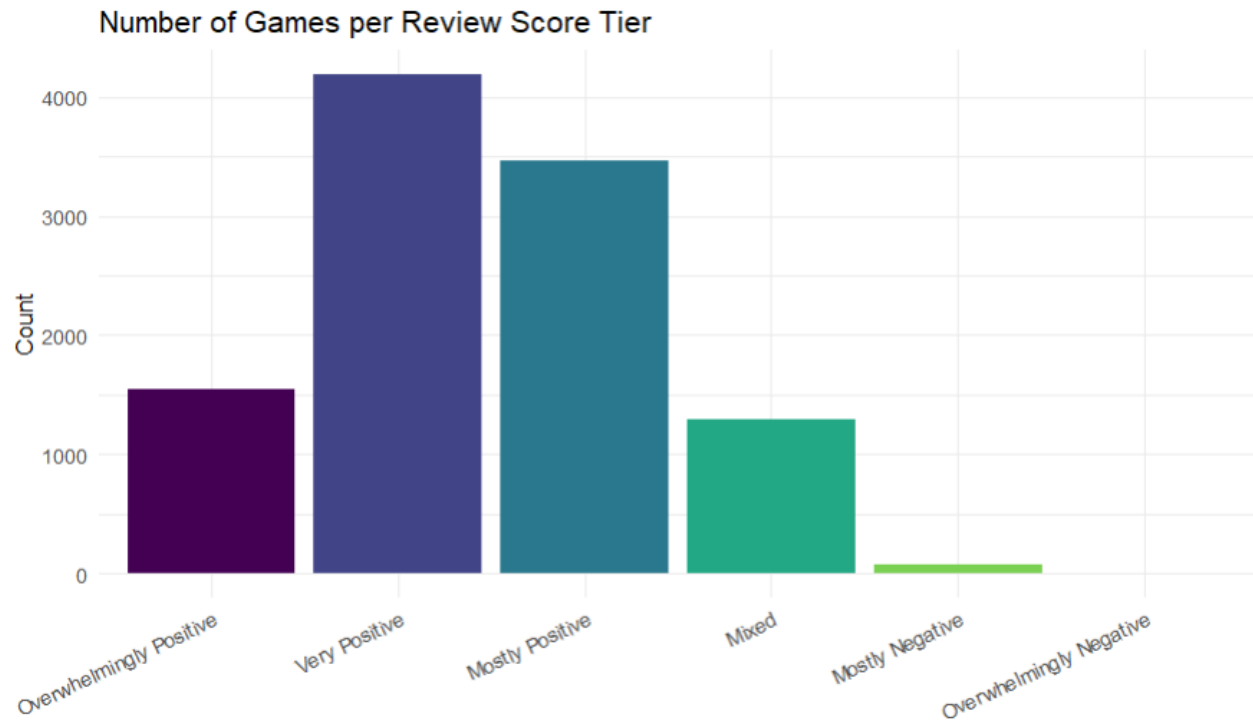
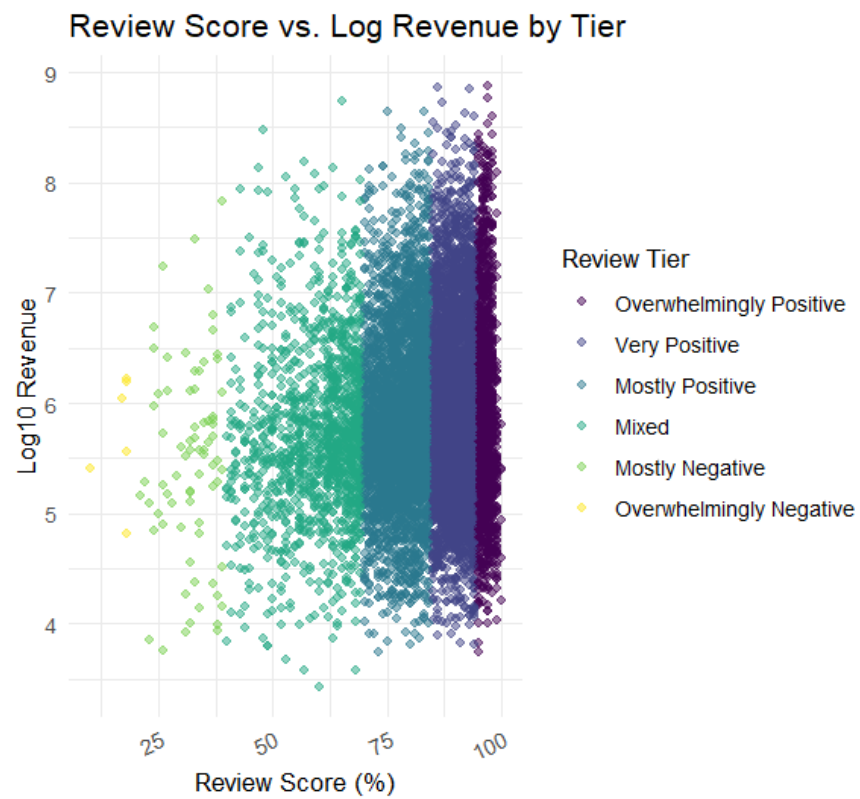


Figure 4.3.4: Scatterplot (x = Review Score, y = Log10 Revenue)



5. Regression Analysis

5.1 Model Setup

Multiple Linear Regression was selected to determine the linear relationship between each input factor and log-transformed revenue while controlling for all other variables.

The initial regression equation is:

K-Fold Cross Validation ($k=5$) was applied using the caret package with `set.seed(123)` to ensure reproducibility. RMSE and R^2 were used as the primary evaluation metrics.

5.2 Initial Model Results

As seen in Table 5.2.1, the initial regression model was run with five input factors on 10,580 observations. The model was statistically significant overall. All five variables were statistically significant at the 0.05 level: `sqrt_price`, `sqrt_reviewScore`, `log_avgPlaytime`, `publisher_AA`, and `publisher_Indie`.

The model produced an R^2 of 0.5053 and an adjusted R^2 of 0.5051, which means approximately 50% of the variation in `log_revenue` was explained by the included predictors. As seen in Table 5.2.2, there is an RMSE of 0.5637 and a R^2 (CV) of 0.5048, which suggests that the model has moderate predictive performance.

5.3 Outlier Detection, Removal, & Factor Elimination

Outliers were identified using standardized residuals computed from the initial model. The threshold applied was ($|\text{standardized (Std.) residuals}| > 3$).

As seen in Table 5.3.1, forty observations were identified as outliers using a standardized residual threshold of $|\text{std. residual}| > 3$. These represent exceptional AAA and Indie game titles rather than data errors. After removing these observations, the sample size decreased from 10,580 to 10,540 observations.

As seen in Table 5.3.2, another regression model was estimated using the cleaned dataset. With the expanded dataset, all five predictors remained statistically significant. However, `sqrt_reviewScore` and `publisher_AA` were removed to maintain a parsimonious model consistent with the original analysis design, keeping the focus on the three most practically interpretable predictors.

5.4 Final Model Results

As seen in Table 5.4.1, the final regression model was run with three input factors on 10,540 observations.

The final regression equation:

The final reduced model remained significant overall, $F(3, 10536) = 3346$, $p < 2.2e-16$, indicating that the retained variables explained a statistically significant portion of the variation in log_revenue. All remaining predictors remained statistically significant.

The reduced model produced an R^2 of 0.4879 and an adjusted R^2 of 0.4878, which means approximately 49% of the variation in log_revenue was explained by the three retained predictors. As seen in Table 5.4.2, there is an RMSE of 0.5691 and a R^2 (CV) of 0.4876. While the reduced model performs slightly below the cleaned five-factor model, it provides a cleaner and more interpretable result with three highly significant predictors.

5.5 Model Performance Comparison

As seen in Table 5.5.1, a comparison of the three models was made. The initial model achieved an R^2 (CV) of 0.5049 with an RMSE of 0.5637. After removing the forty outliers, both R^2 (CV) improved to 0.5163 and RMSE improved to 0.5529. The final reduced model, using only three predictors, produced an R^2 (CV) of 0.4876 and RMSE of 0.5691, a modest trade-off in performance for a significantly simpler and more interpretable model.

Since the final reduced model provides strong predictive performance using fewer variables, it was selected as the preferred model.

5.6 Interpretation

The final reduced model contains three statistically significant predictors of log_revenue:

sqrt_price ($\beta = 0.243$, $p < 2e-16$) is the strongest predictor in the final model. For every one unit increase in sqrt_price, log_revenue increases by 0.243 holding all other factors constant. This validates the price tier dimension findings and directly addresses the concern that price effects may be driven entirely by publisher class, price has a measurable independent contribution to revenue beyond simply being a proxy for AAA publisher status.

log_avgPlaytime ($\beta = 0.553$, $p < 2e-16$) is the second strongest predictor. For every one unit increase in log_avgPlaytime, log_revenue increases by 0.553 holding all other factors constant. Games that sustain long-term player engagement are significantly associated with higher revenue, likely because engaging games drive continued sales through word-of-mouth, positive reviews, and sustained platform visibility.

publisher_Indie ($\beta = -0.359$, $p < 2e-16$) indicates that Indie games earn significantly less revenue than AAA games after controlling for price and playtime. This confirms that publisher

class independently influences revenue outcomes, AAA publishers benefit from marketing scale, brand recognition, and distribution advantages that Indie developers cannot replicate regardless of price point or game quality.

The model's R^2 of approximately 49% indicates that the selected input factors explain a substantial portion of the variance in log_revenue, a significant improvement over the original 31% achieved with the smaller dataset. The remaining 51% of variance is unexplained by the available predictors, suggesting that revenue is also influenced by factors outside the scope of this dataset such as marketing spend, franchise recognition, and platform visibility. Forty outliers representing genuine exceptional game titles were removed prior to final modeling. See Section 2.4 for a full discussion of dataset scope and limitations.

5.7 Visualizations

Table 5.2.1: Initial Model Coefficients

Variable	Estimate	Std. Error	t value	p value
Intercept	5.239	0.034	152.68	< 2e-16 ***
sqrt_price	0.243	0.005	52.48	< 2e-16 ***
sqrt_reviewScore	-0.093	0.004	-23.06	< 2e-16 ***
log_avdPlaytime	0.506	0.018	28.79	< 2e-16 ***
publisher_AA	-0.265	0.216	-12.26	< 2e-16 ***
publisher_Indie	-0.575	0.020	-28.77	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 5.2.2: Initial Model Performance

Metric	Value
R^2	0.5074
Adjusted R^2	0.5072
RMSE (CV)	0.5625
R^2 (CV)	0.5071
Residual Std. Error	0.0077
F-statistic	2176 on 5 and 10,562 DF
p value	< 2.2e-16

Table 5.3.1: Identified Outliers (head: only first 6 rows)

Row	log_rev enue	sqrt_pri ce	sqrt_re viewSc ore	log_av gPlayti me	publish er_AA	publish er_Indi e	Predict ed	Std. Redisu al
7	8.6394	3.8717	4.2426	1.8928	0	0	6.7477	3.368
11	8.5975	3.1607	1.7320	1.5142	0	0	6.6151	3.529
20	8.4502	5.4763	3.6056	1.3609	1	0	6.6634	3.179
21	8.4381	3.8717	1.7321	1.8274	1	0	6.6819	3.126
27	8.3798	4.4710	2.4495	1.5683	1	0	6.6304	3.112
30	8.3571	4.4710	2.8284	1.4300	1	0	6.5253	3.259

Table 5.3.2: Outliers Removed Model Coefficients

Variable	Estimate	Std. Error	t value	p value
Intercept	5.290	0.034	154.17	< 2e-16 ***
sqrt_price	0.248	0.005	54.17	< 2e-16 ***
sqrt_reviewScore	-0.091	0.004	-23	< 2e-16 ***
log_avdPlaytime	0.506	0.017	29.14	< 2e-16 ***
publisher_AA	-0.270	0.021	-12.67	< 2e-16 ***
publisher_Indie	-0.571	0.020	-29.01	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 5.3.3: Outliers Removed Model Performance

Metric	Value
R ²	0.5186
Adjusted R ²	0.5184
RMSE (CV)	0.5521
R ² (CV)	0.5178
Residual Std. Error	0.5519
F-statistic	2267 on 5 and 10523 DF
p value	< 2.2e-16

Table 5.4.1: Final Model Coefficients

Variable	Estimate	Std. Error	t value	p value
Intercept	4.615	0.025	188.29	< 2e-16 ***
sqrt_price	0.243	0.005	51.99	< 2e-16 ***
log_avdPlaytime	0.553	0.018	31.04	< 2e-16 ***
publisher_Indie	-0.358	0.013	-27.32	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 5.4.2: Final Model Performance

Metric	Value
R ²	0.4878
Adjusted R ²	0.4876
RMSE (CV)	0.5695
R ² (CV)	0.4870
Residual Std. Error	0.5692
F-statistic	3341 on 3 and 10525 DF
p value	< 2.2e-16

Table 5.5.1: Model Performance Comparison

Model	n	Factors	R ² (CV)	Adj. R ²	RMSE (CV)
Initial Model	10,580	5	0.5071	0.5051	0.5625
Outliers Removed	10,580	5	0.5178	0.5165	0.5521
Final Model	10,580	3	0.487	0.4878	0.5695

Figure 5.2.1: Actual vs Predicted Log Revenue (Initial Model)

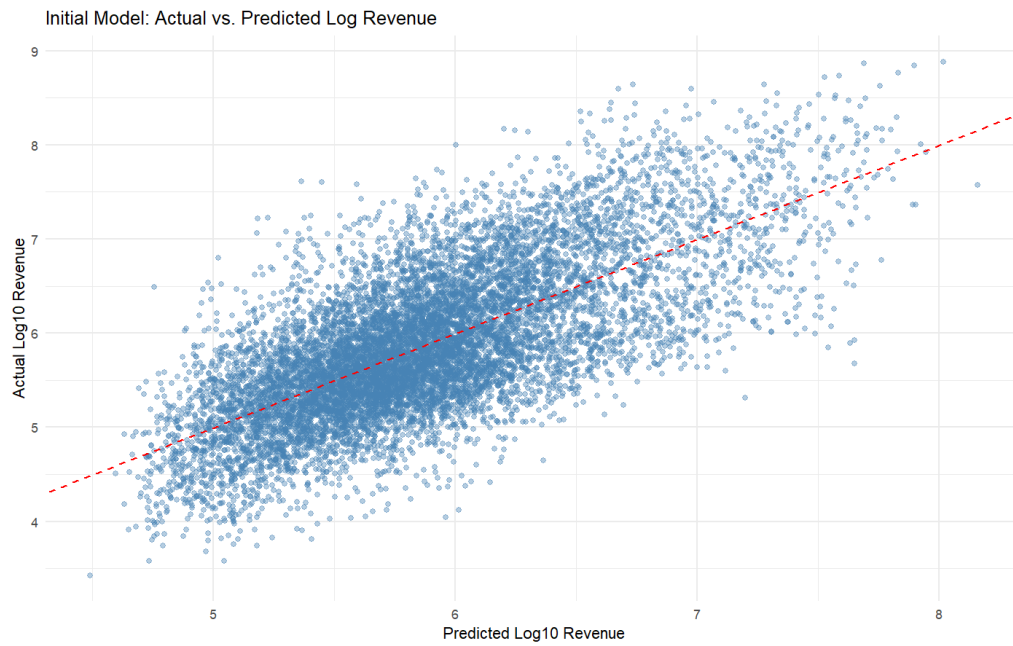
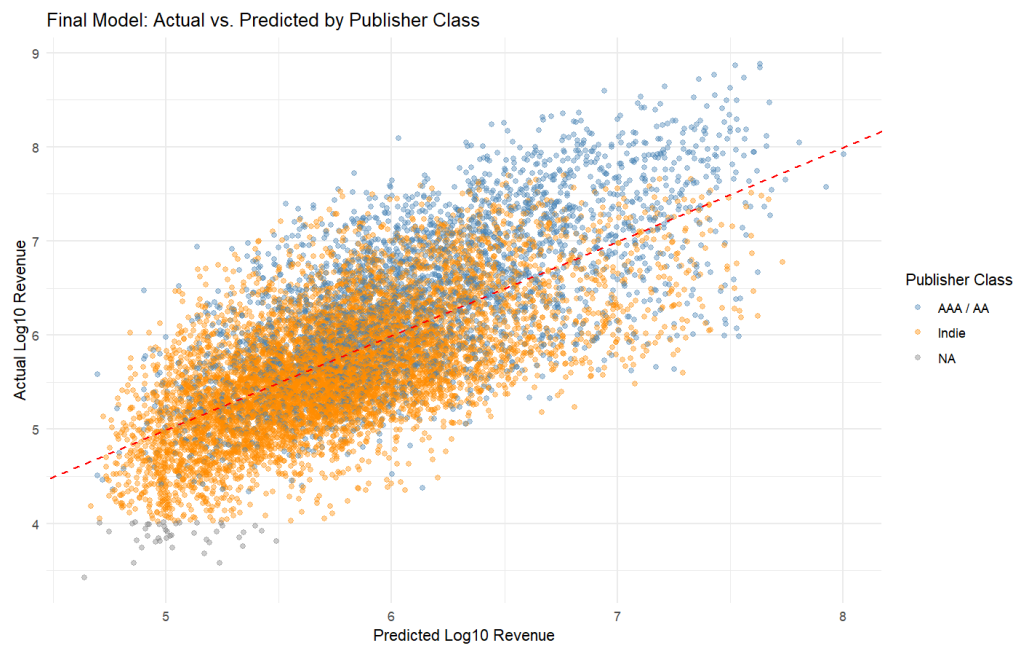


Figure 5.4.1: Final Reduced Model: Actual vs Predicted by Publisher Class



6. Random Forest Analysis

6.1 Model Setup

For the machine learning portion of the project, a Random Forest regression model was used to predict log-transformed revenue for Steam games. The target variable was `log_revenue` instead of raw revenue because revenue was highly skewed and included extreme high-earning titles. Using log revenue reduced the influence of those outliers and made the model results more stable.

The Random Forest was used as the overall project-level predictive model, not just for Dimension 1. Hobbyist titles were excluded so the model aligned with the final AAA, AA, and Indie analysis scope. The final predictor set included `publisherClass`, `price_tier`, `review_tier`, `price`, `reviewScore`, `avgPlaytime`, `review_count`, and `wishlists`.

Direct revenue-related variables were excluded to avoid data leakage. This included `revenue`, `copiesSold`, `log_copiesSold`, `players_total`, `total_revenue`, M1/Y1 revenue and sales, and yearly revenue/sales fields. These variables are too directly connected to the outcome and would make the model appear more accurate than it would be in a real predictive setting.

The model used 10,572 complete AAA, AA, and Indie observations. The data was split into an 80% training set and a 20% testing set, stratified by publisher class, producing 8,487 training observations and 2,085 testing observations. The final Random Forest used 500 trees with `mtry = 2` and performed strongly, with a test-set R-squared of 0.907, RMSE of 0.242, and MAE of 0.189.

6.2 Feature Importance Results

The Random Forest model identified `review_count` as the strongest predictor of `log_revenue`. This suggests that market attention and player response volume are major signals of revenue performance; games with more review activity tend to have stronger revenue outcomes.

The next strongest predictors were `price`, `wishlists`, and `avgPlaytime`. This shows that revenue performance is shaped by pricing power, pre-purchase demand, and player engagement. Average playtime still matters, but it was not the top predictor in the updated model.

`reviewScore` also contributed, but it ranked below review count, price, wishlists, and average playtime. This suggests that the volume of player response is more predictive of revenue than review score alone.

Publisher class, price tier, and review tier were still useful predictors, but they ranked lower than the direct demand and engagement variables. Overall, the model relied most on review count, price, wishlists, and playtime when predicting log revenue.

Final variable importance ranking:

1. Review count
2. Wishlists
3. Price
4. Average playtime
5. Review score
6. Publisher class
7. Price tier

8. Review tier

Note: Categorical fields are reported at the variable level, such as `publisherClass`, `price_tier`, and `review_tier`, rather than as separate dummy categories.

6.3 Model Performance

The final Random Forest regression model was evaluated on the 20% testing set after excluding Hobbyist titles. The model produced an RMSE of 0.242, an MAE of 0.189, and an R-squared of 0.907 for predicting `log_revenue`.

This R-squared value means the model explained about 90.7% of the variation in log revenue on the test set, indicating strong predictive performance. This result is likely driven by strong demand and engagement signals, especially `review_count`, `price`, `wishlists`, and `avgPlaytime`.

Visual diagnostics support this result. The Actual vs. Predicted plot shows a strong positive relationship between actual and predicted log revenue, meaning the model captured the overall revenue pattern well. The residual plot shows that most prediction errors cluster around zero, with some larger errors remaining for harder-to-predict games.

Overall, the Random Forest model performed strongly as a project-level predictive model. However, it should still be interpreted as predictive rather than causal: it identifies variables associated with revenue, but does not prove that any single factor directly causes higher revenue.

6.4 Comparison with Regression Model

While the regression analysis examined direct relationships between variables, the Random Forest model expanded the analysis by adding non-parametric flexibility. Steam revenue is multidimensional, and financial performance is rarely explained by one linear factor. Random Forest is useful because it can capture nonlinear patterns and interactions without requiring those relationships to be specified in advance.

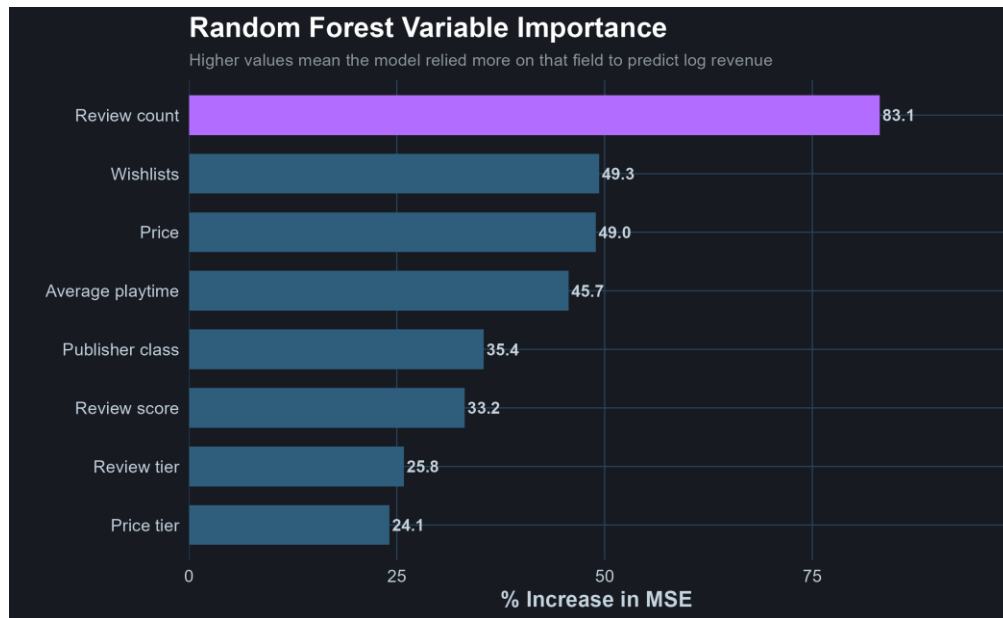
This matters because market factors work together. For example, publisher class, price, review activity, wishlists, and playtime may interact differently across games. A premium AAA game and a lower-priced Indie game can follow very different revenue paths even when both have strong review scores.

The Random Forest model also provided a feature importance ranking. In the updated whole-dataset model, `review_count` was the strongest predictor, followed by `wishlists`, `price`, and `avgPlaytime`. This supports the broader finding that demand signals, pricing power, and engagement are stronger predictors of revenue than review score alone.

The Random Forest model does not replace the regression analysis. Instead, it complements it by showing which variables improve prediction when the project's major market dimensions are considered together.

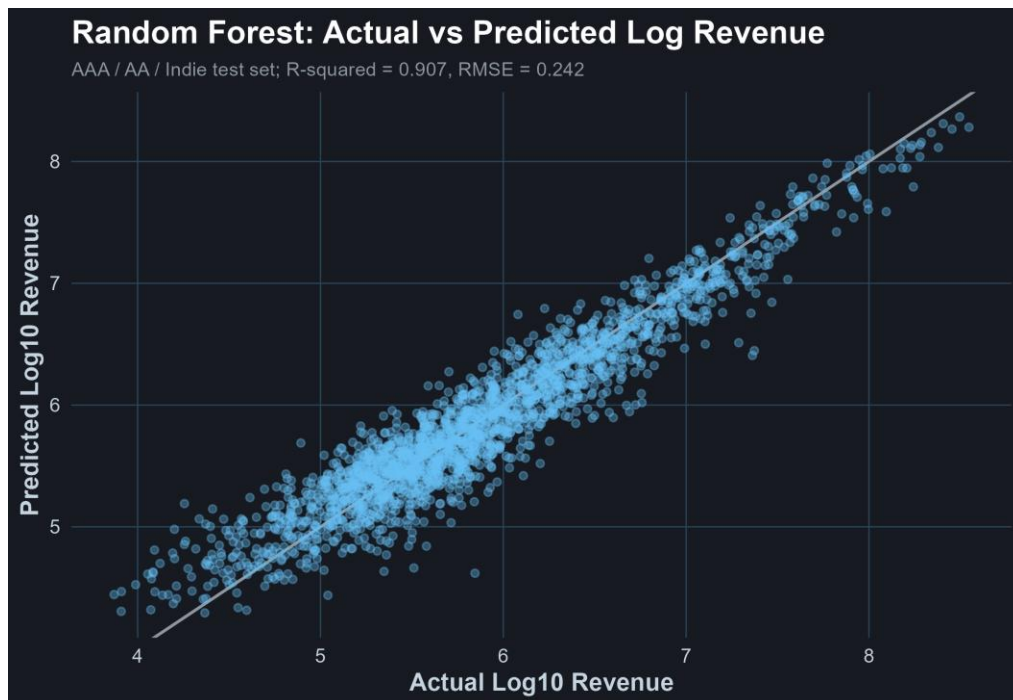
6.5 Visualizations

The Random Forest section includes three main visuals. The first is the variable importance chart, which shows that average playtime, price, publisher class, and price tier were the strongest predictors of log revenue. This chart is the main visual for explaining which factors mattered most to the model.



Variable Importance:

The Random Forest variable importance chart shows that `review_count` was the strongest predictor of log revenue, followed by `wishlists`, `price`, and `average playtime`. This suggests that market attention, demand, pricing power, and engagement were stronger revenue signals than review score or category-based variables alone.



Actual vs. Predicted:

The Actual vs. Predicted Log Revenue plot shows a strong fit between the model's predictions and the actual test-set values, with most points clustering closely around the diagonal reference line. The model explains a large share of variation in log revenue ($R\text{-squared} = 0.907$), though some games still show larger prediction errors.



Residual Plot:

The residual plot shows that most prediction errors cluster around zero, meaning the model generally predicted log revenue without strong systematic bias. A few larger residuals remain, but the overall pattern supports a stable model fit across the AAA, AA, and Indie test set.

7. Recommendations & Conclusions

7.1 Key Findings Summary

Dimension 1 showed that publisher class is strongly associated with revenue performance. Indie games dominate the dataset by volume, making up 70.3% of analyzed titles, but they capture only 19.5% of total revenue. AAA games represent only 9.7% of titles but capture 44.3% of revenue and have the highest median revenue at approximately \$5.14M.

The statistical results support this pattern. Welch ANOVA found significant differences in mean log revenue across publisher classes, $F = 1095.061$, $p < 0.001$. Pairwise comparisons showed significant differences between AAA, AA, and Indie games. The adjusted regression also showed that publisher class remains meaningful after controlling for price, review score, and average playtime.

Dimension 2 showed the analysis confirms that price tier is a statistically significant predictor of revenue among Steam games in this dataset. Premium pricing (\$40+) is most strongly associated with higher net revenue, with all three tier comparisons showing statistically significant differences. The budget tier (Under \$20), despite dominating the dataset in volume at 75% of games, consistently underperforms the other tiers, suggesting that competing on low price alone is the least effective pricing strategy.

Dimension 3 showed that review score tier has a statistically significant but practically modest relationship with net revenue. The one-way ANOVA found significant differences in mean log revenue across review score tiers ($F = 20.63$, $p < 2e-16$). However, mean log revenue rises only gradually across tiers, from 5.72 (Mixed) to 5.93 (Very Positive), with the most meaningful separations occurring between lower tiers and the top tiers rather than between adjacent positive tiers. Post-hoc Tukey testing confirmed that Very Positive vs. Overwhelmingly Positive was not statistically significant ($\text{diff} = -0.0096$, $p = 0.9986$), meaning the two highest review tiers produce essentially equivalent revenue outcomes. The correlation matrix reinforced this, showing a near-zero linear relationship between review score and log revenue ($r = -0.1005$). The Overwhelmingly Positive tier led in mean revenue at \$9.50M, but its median of \$649,204 reveals this is driven by blockbuster outliers rather than broad success across the tier. Overall, review score alone is a weak and unreliable predictor of revenue compared to price and playtime.

The Regression analysis confirms that price, playtime, and publisher class are statistically significant predictors of log transformed revenue among Steam games in the dataset. The final reduced model explained approx. 49% of the variance in revenue using three predictors, with `log_avgPlaytime` emerging as the strongest positive predictor. `sqrt_price` and `publisher_Indie` are significant negative predictors which confirms that Indie games earn less than AAA games even after controlling for price and playtime.

The Random Forest model performed strongly as the project-level predictive model. After excluding Hobbyist titles and leakage-related variables, the model used 10,572 complete AAA, AA, and Indie observations and achieved $R^2 = 0.906$, $\text{RMSE} = 0.242$, and $\text{MAE} = 0.189$ on the test set. The strongest predictor was review count, followed by wishlists, price, and average playtime. This shows that revenue prediction was driven most by market attention, demand, pricing, and engagement. Publisher class, review score, review tier, and price tier also contributed, but ranked below direct demand and engagement signals.

7.2 Business Recommendations

Overall, revenue performance is shaped by market demand, engagement, pricing, and publisher position. Review count, wishlists, and average playtime were among the strongest Random Forest predictors, suggesting that games benefit from visibility, audience interest, and sustained player engagement.

Pricing also matters, but low price alone does not guarantee stronger revenue. Higher price tiers are associated with stronger revenue outcomes, though these tiers are often supported by larger publisher resources and stronger market visibility. For Indie developers, the findings suggest that competing only on low price may limit revenue potential; stronger positioning, wishlist growth, community activity, and engagement may be more valuable.

Publisher class remains an important revenue signal. AAA games capture the largest revenue share despite representing a small portion of the dataset, while Indie games dominate title count but earn a smaller share of total revenue. Review score is still useful, but it is less predictive than review volume, wishlists, price, and playtime. The strongest recommendation is to prioritize market visibility and player engagement alongside pricing strategy.

9. References

Gamalytic. (2023). *Gamalytic: Steam game analytics and market research*.
<https://gamalytic.com/about>